Selin Choi

Ph.D. student in Culture Technology · KAIST

UVR Lab · Advised by Prof. Woontack Woo · Daejeon, South Korea

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I study spatial recording and adaptive playback in AR/VR, with a focus on scene graphs and task-aware navigation.

Education

2025 - Present
Ph.D. student in Culture Technology
KAIST · UVR Lab

2023 - 2025
M.S. in Culture Technology
KAIST · UVR Lab

2018 - 2023
B.S. in Software
Kwangwoon University
Major in System Software · GPA 4.23 / 4.5

Research experience

Jun - Aug 2026
Monash University
Visiting Researcher · Embodied Visualisation Group
Host: Prof. Tim Dwyer
Immersive analytics and AR/VR data visualisation.

Jan - Feb 2022
ETRI · VR/AR Content Lab
Winter Research Intern
AR test program development for traditional buildings in the H-BIM project.

Research projects & funding

2026 · Selected
Doctoral Research Grant
AI-Driven Learner-Adaptive Spatial Experience Playback Framework in AR/VR Environments
National Research Foundation of Korea (NRF)

Apr 2024 - Present
MetaObject
Participating researcher
Real-time XR Interface Technology Development for Environmental Adaptation
Intelligent Platform Team · IITP, Ministry of Science and ICT

Mar - Dec 2023
Dynamic Digital Twin for XR Collaboration
Participating researcher

Awards & scholarships

2024
Best Conference Paper Award · 1st Prize
IEEE International Symposium on Mixed and Augmented Reality

2024
Outstanding Paper Award
Korean Institute of Information Scientists and Engineers

2023
2nd Place · Metaverse Developer Contest
K-META

2023 -
Government Scholarship
Fully funded M.S. and Ph.D. studies at KAIST

Teaching · KAIST

Spring 2026
3D Interaction Design
GCT555

Fall 2025
Fall 2026
Augmented Reality Project
GCT600 / MV600

Publications

International journals & conferences

2026 **Object-Driven Event Retrieval with Assembled Object Interface for Mixed Reality Assembly Playback**
Selin Choi, Sun Ah Bae, Dooyoung Kim, and Woontack Woo
IEEE ISMAR 2026 / Paper

2026 **Task Breakpoint Generation using Origin-Centric Graph in Virtual Reality Recordings for Adaptive Playback**
Selin Choi, Dooyoung Kim, Taewook Ha, Seonji Kim, and Woontack Woo
IEEE Transactions on Visualization and Computer Graphics / Paper

2026 **Spatial Affordance-Aware Affine Transformation Between Heterogeneous Spaces for Mixed Reality Remote Collaboration**
Seonji Kim, Dooyoung Kim, **Selin Choi**, and Woontack Woo
IEEE Transactions on Visualization and Computer Graphics / Paper

2025 **Meta-Objects: Interactive and Multisensory Virtual Objects Learned From the Real World for Use in Augmented Reality**
Dooyoung Kim, Taewook Ha, Jinseok Hong, Seonji Kim, **Selin Choi**, Heejeong Ko, and Woontack Woo
IEEE Computer Graphics and Applications / Paper

2025 **Human-Scene Interaction Data Generation with Virtual Environment using User-Centric Scene Graph**
Taewook Ha, **Selin Choi**, Seonji Kim, Dooyoung Kim, and Woontack Woo
IEEE VR Abstracts and Workshops (VRW) / Paper

2024 **Spatial Affordance-Aware Interactable Subspace Allocation for Mixed Reality Telepresence**
Dooyoung Kim, Seonji Kim, **Selin Choi**, and Woontack Woo
IEEE ISMAR 2024 / Paper
Best Conference Paper · 1st Prize

Domestic conferences

2024 **Undo Slider for Multi-Step Editing in Collaborative AR Environments**
Selin Choi, Minju Baeck, and Woontack Woo
KIISE · pp. 31-33

2024 **Virtual Consumable Resources Interacting with Real-World Tools in a Remote XR Training System**
Selin Choi, Yoonjae Hong, Minji Park, and Woontack Woo
KIISE · pp. 248-250
Outstanding Paper Award

2023 **Interactive 3D Voxel Drawing System Using Bare-Hand Interaction with Augmented Reality Glasses**
Selin Choi, Minju Baeck, Hyungwoo Jin, Taewook Ha, and Woontack Woo
KIISE

Tools & technologies Unity · C# · C++ · Swift · Python · Xcode · Figma